

APEXPLANET SOFTWARE PVT. LTD.

Cybersecurity & Ethical Hacking

Internship

TASK 2 — NMAP SCAN REPORT

Network & Service Enumeration

- **Objective:**

1. The objective of this activity was to perform network reconnaissance and service enumeration against the Metasploitable 2 target using Nmap.
2. The scan was performed from the Kali Linux attacker machine within the isolated Host-Only network established during Task 1. The assessment focused on identifying reachable TCP ports, exposed network services, service and version information, operating system details, and UDP services.

- **Lab Environment:**

1. The Nmap assessment was performed within the controlled cybersecurity laboratory environment configured using Oracle VirtualBox. Kali Linux was used as the attacker and scanning machine, while Metasploitable 2 was used as the intentionally vulnerable target.
2. Both virtual machines were connected through the private Host-Only network, allowing network scanning to be performed in an isolated environment.

Component	Configuration
Virtualization	Oracle VirtualBox
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2
Network Type	Host-Only Adapter
Kali IP	192.168.56.101
Target IP	192.168.56.102
Scanning Tool	Nmap

Scanning Methodology

The assessment was performed in multiple stages to progressively gather information about the target.

Stage 1 — TCP Port Enumeration

An initial Nmap scan was performed to identify open TCP ports and determine which network services were exposed by the target.

```
nmap -Pn 192.168.56.102
```

Stage 2 — Service and Version Detection

Nmap service detection was then used to identify the applications and versions associated with the discovered services.

```
nmap -Pn -sV 192.168.56.102
```

Stage 3 — Operating System Detection

The scan results were also used to identify the operating-system characteristics of the target.

Stage 4 — UDP Enumeration

A UDP scan of the most common 20 UDP ports was performed to identify additional UDP services.

```
sudo nmap -Pn -sU --top-ports 20 192.168.56.102
```

• TCP Port Enumeration:

1. Initial TCP Scan

An initial TCP scan was performed from Kali Linux against the Metasploitable 2 target using:

```
nmap -Pn 192.168.56.102
```

The scan confirmed that the target was active and identified **23 open TCP ports**, while **977 TCP ports were reported as closed**.

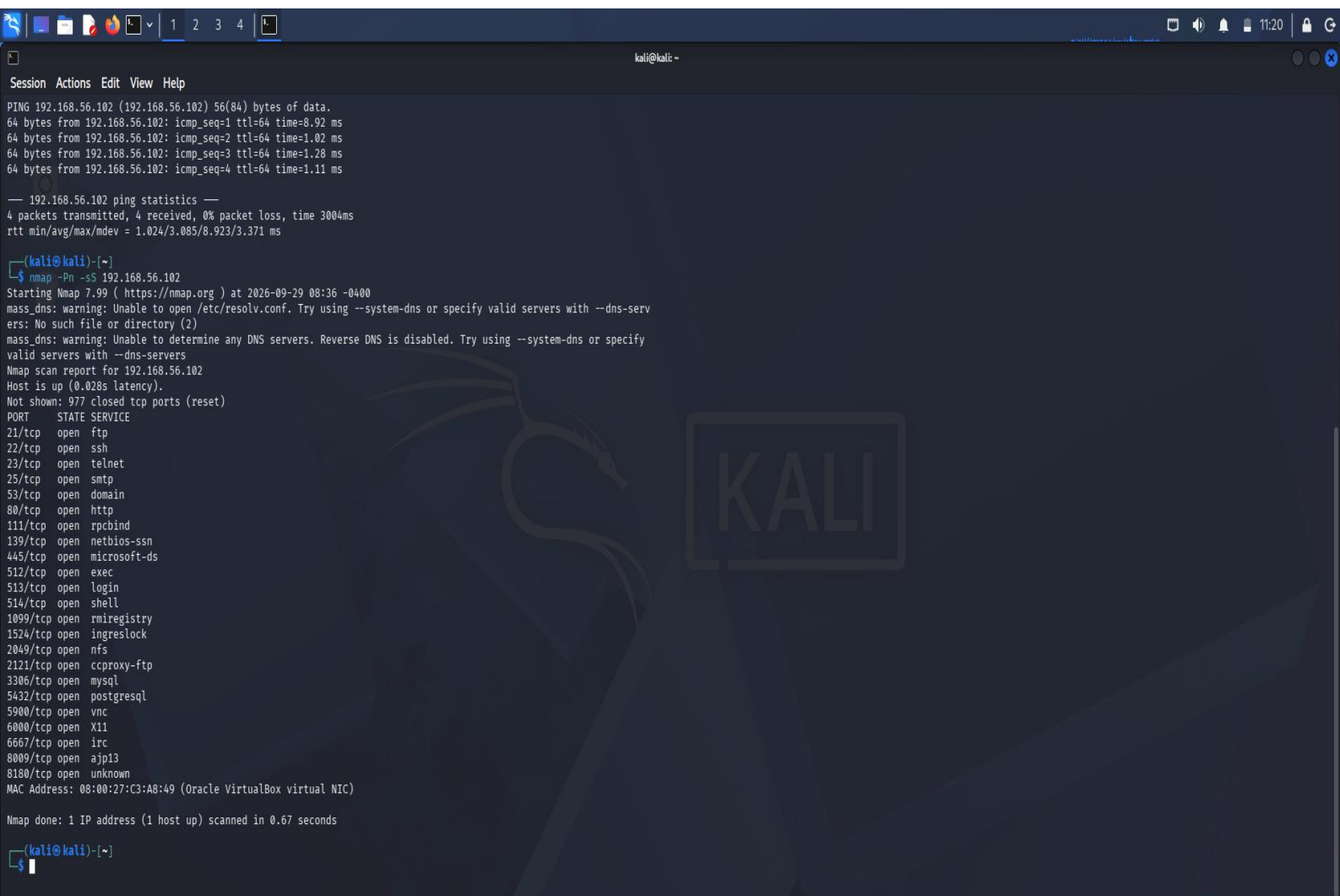
2. Scan Results

The scan identified a wide range of exposed services, including FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, NFS, MySQL, PostgreSQL, VNC, X11, IRC, AJP13 and other network services.

3. Observation

The large number and variety of exposed services indicate that the target has a broad network-facing attack surface. Remote-access, file-sharing, database and web services were all accessible from the scanning machine. These results provide the initial service map of the target. Service and version detection was subsequently performed to identify the specific software running behind the discovered ports.

Figure 1: Nmap TCP port scan results



```
Session Actions Edit View Help
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=8.92 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.02 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.28 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=1.11 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.024/3.085/8.923/3.371 ms

(kali@kali)~$ nmap -Pn -sS 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-29 08:36 -0400
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-serv
ers: No such file or directory (2)
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify
valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.028s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:C3:A8:49 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds

(kali@kali)~$
```

• **Service & Version Detection:**

1.Scan Performed

Nmap service and version detection was performed against the target using:

```
nmap -Pn -sV 192.168.56.102
```

The scan provided detailed information about the applications and services running on the previously identified open TCP ports.

2. Scan Results

The scan identified several services and their associated software versions, including:

- **FTP:** vsftpd 2.3.4
- **SSH:** OpenSSH 4.7p1
- **DNS:** ISC BIND 9.4.2
- **HTTP:** Apache httpd 2.2.8
- **SMB:** Samba
- **MySQL:** 5.0.51a
- **PostgreSQL:** 8.3.x
- **VNC:** Protocol 3.3
- **IRC:** UnrealIRCd

- **AJP13:** Apache Jserv
- **HTTP:** Apache Tomcat/Coyote

3. Observation

Service and version detection provided a more detailed view of the target than basic port enumeration. The identified software versions can be used to research known security issues and determine which services require further vulnerability assessment.

The results from this scan therefore provide an important input for the subsequent OpenVAS vulnerability assessment.

```

kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap -Pn -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-29 11:52 -0400
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers: No such file or directory (2)
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 47.83% done; ETC: 11:53 (0:00:07 remaining)
Nmap scan report for 192.168.56.102
Host is up (0.023s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          netkit-rsh rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:C3:A8:49 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.24 seconds

(kali@kali)-[~]
$

```

Figure 2: Nmap service and version detection results

- **Operating System & UDP Enumeration:**

1. Operating System Detection

Nmap was used to identify the operating-system characteristics of the Metasploitable 2 target. The scan identified the target as a **Linux 2.6.x** system, with the detailed result indicating a Linux 2.6.9–2.6.33 range.

2. UDP Scan

A UDP scan was performed against the most common 20 UDP ports using:

```
sudo nmap -Pn -sU --top-ports 20 192.168.56.102
```

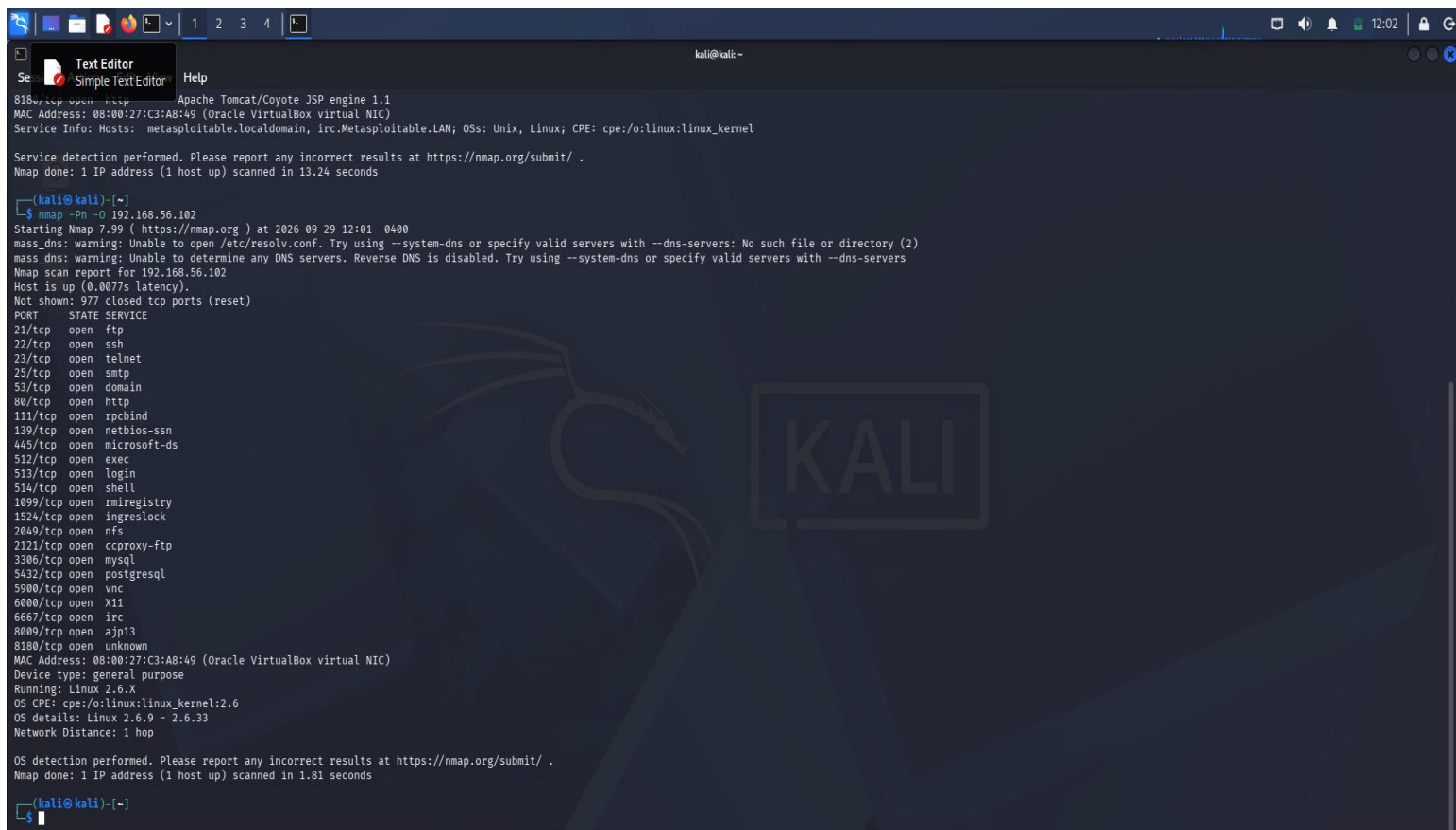
The scan identified several UDP ports as **open or filtered**, including services associated with DNS, DHCP, TFTP, NTP, NetBIOS, SNMP, routing, IPP, and other network protocols.

3. Observation

The OS detection provided additional information about the target platform, while the UDP scan expanded the assessment beyond TCP-based services. The open|filtered results indicate that Nmap could not definitively distinguish between an open port and one filtered by packet filtering.

Together, these scans provide a broader view of the target's network exposure and complement the TCP service enumeration performed earlier.

Figure 3: Nmap operating system detection results

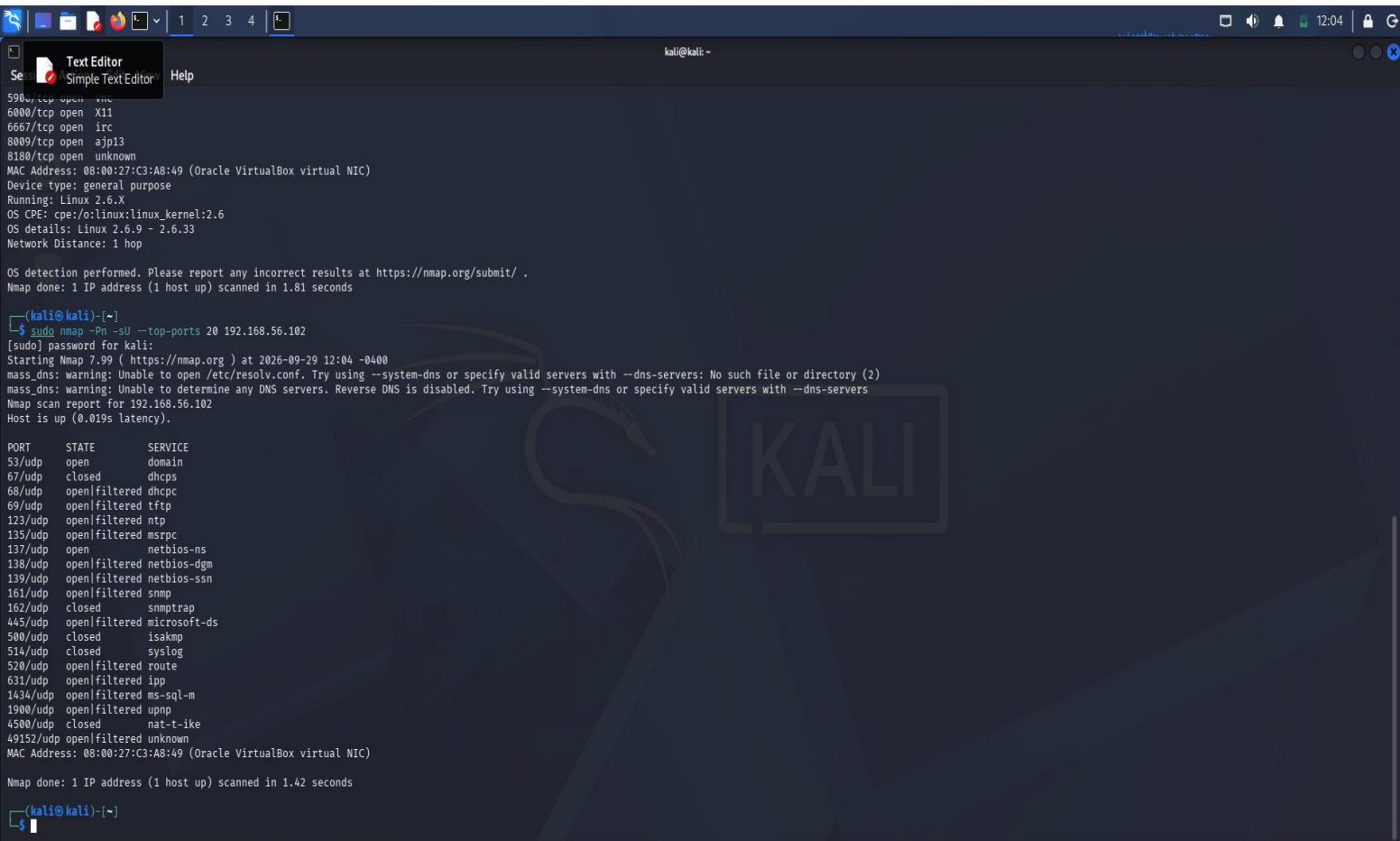


```
(kali@kali)-[~]
$ nmap -Pn -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-29 12:01 -0400
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers: No such file or directory (2)
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.0077s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8080/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:C3:A8:49 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.81 seconds

(kali@kali)-[~]
$
```

Figure 4: Nmap UDP scan results



```
kali@kali: ~  
$ sudo nmap -Pn -sU --top-ports 20 192.168.56.102  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-29 12:04 -0400  
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers: No such file or directory (2)  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.102  
Host is up (0.019s latency).  
  
PORT      STATE      SERVICE  
53/udp    open       domain  
67/udp    closed     dhcp  
68/udp    open|filtered dhcpc  
69/udp    open|filtered tftp  
123/udp   open|filtered ntp  
135/udp   open|filtered msrpc  
137/udp   open       netbios-ns  
138/udp   open|filtered netbios-dgm  
139/udp   open|filtered netbios-ssn  
161/udp   open|filtered snmp  
162/udp   closed     snmptrap  
445/udp   open|filtered microsoft-ds  
500/udp   closed     isakmp  
514/udp   closed     syslog  
520/udp   open|filtered route  
631/udp   open|filtered ipp  
1434/udp  open|filtered ms-sql-m  
1900/udp  open|filtered upnp  
4500/udp  closed     nat-t-ike  
49152/udp open|filtered unknown  
MAC Address: 08:00:27:C3:A8:49 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.42 seconds  
  
kali@kali: ~
```

• Detailed Analysis

The Nmap assessment provided a multi-layered view of the Metasploitable 2 target. The initial TCP scan identified **23 open TCP ports**, followed by service and version detection that revealed the applications operating on those ports.

The scan identified several categories of exposed services, including:

1. Remote-access services such as SSH and Telnet
2. File-transfer services such as FTP
3. Web services including Apache and Tomcat
4. File-sharing services such as SMB and NFS
5. Database services including MySQL and PostgreSQL
6. Remote graphical services such as VNC and X11
7. Additional services such as DNS, IRC and AJP13

The UDP scan further identified several ports as open|filtered, showing that the target's network exposure was not limited to TCP services.

• Security Observations

The enumeration demonstrates that the target exposes a broad range of network services. The identified service versions provide useful information for subsequent security assessment and vulnerability research.

However, **Nmap enumeration alone does not confirm that these services are vulnerable**. The identified services and versions will therefore be assessed further using OpenVAS.

• Results

Activity	Result
TCP port enumeration	23 open TCP ports identified
Service detection	Successfully completed
Version detection	Successfully completed
OS detection	Linux 2.6.x identified
UDP enumeration	Successfully completed
Attack-surface mapping	Successfully completed

• **Conclusion**

The Nmap assessment was successfully completed against the Metasploitable 2 target. The scans identified exposed TCP and UDP services, application information, and operating-system characteristics.

The resulting service map provides a baseline for the next stage of the assessment, where OpenVAS will be used to identify and analyze vulnerabilities associated with the discovered services.